Sampling and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

One connected survey-plan record, with a graph extension.

Define population, label a result, repair a claim and retain nonresponse before designing a fresh plan.

Keep one habit: Keep question, frame, selection method, invited/responded counts and limits together.

MAIN LESSON · SECTION 1

Recall · Retrieve earlier ideas

> Try before opening notes.

Your evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Missing Data

Overfitting Intuition

Types of Classification Error

W1 · Define the target population

> Give a sample whose selection cannot support that whole-population claim.

Your evidence record

- Main sample reference: population means everyone the question concerns; frame is the source from which invitations are selected; respondents are those who answer. Original club code counts2 Code choices of3 responses. W2 is a separate longer-lunch result7 of12. Pictured format plan uses60 population,15 invited,12 replies; the fraction12/15 is not12/60. No preference result is supplied for that plan.

- The question is whether all Grade 7 students prefer a longer lunch. Name the **population** and one unsuitable **sample**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Label the observed lunch proportion

> Keep the12-respondent denominator and avoid a whole-grade claim.

Your evidence record

- Main sample reference: population means everyone the question concerns; frame is the source from which invitations are selected; respondents are those who answer. Original club code counts2 Code choices of3 responses. W2 is a separate longer-lunch result7 of12. Pictured format plan uses60 population,15 invited,12 replies; the fraction12/15 is not12/60. No preference result is supplied for that plan.

- Seven of 12 fictional respondents prefer longer lunch. State the result without generalizing.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair a club-only headline

> Narrow the claim or propose a changed sampling plan.

Your evidence record

- Main sample reference: population means everyone the question concerns; frame is the source from which invitations are selected; respondents are those who answer. Original club code counts2 Code choices of3 responses. W2 is a separate longer-lunch result7 of12. Pictured format plan uses60 population,15 invited,12 replies; the fraction12/15 is not12/60. No preference result is supplied for that plan.

- A report says all Grade 7 prefers coding after surveying only the coding club. Rewrite it.

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Report nonresponse after fair selection

> Fair invitations do not reveal the views of nonrespondents.

Your evidence record

- Main sample reference: population means everyone the question concerns; frame is the source from which invitations are selected; respondents are those who answer. Original club code counts2 Code choices of3 responses. W2 is a separate longer-lunch result7 of12. Pictured format plan uses60 population,15 invited,12 replies; the fraction12/15 is not12/60. No preference result is supplied for that plan.

- Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report non**response**?

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Plan independently

> Keep the first attempt before hints.

Your evidence record

- Use a fictional grade and invented identifiers. Design the sampling/anonymous-response process; no real roster or real survey is required.
- Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.
- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

> **My independent explanation, trace and limits**

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Plan independently

MAIN LESSON · SECTION 9

Project · Keep the decision and evidence

> Extend the existing project record.

Your evidence record

- Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.
- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 22

Feedback · Repair and return

> Keep first attempt, repair and a changed-case check.

Your evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	

Evidence	My record
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Point to work that supports your judgement.

Your evidence record

- - I can explain the mechanism.
  - I can show evidence.
  - I can handle a changed case.
  - I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use W2: Read the **sample** result for Questions **population**s and **sample**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair a claim

REFERENCE · SECTION 3

Extra practice · Repair a claim

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier target

REFERENCE · SECTION 5

Reference · Earlier target

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

▼ **Extra practice · Earlier worked case**

REFERENCE · SECTION 6

Extra practice · Earlier worked case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · Changed case**

REFERENCE · SECTION 7

Extra practice · Changed case

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My practice reasoning

> **Write or draw your reasoning and evidence.**

Week 19 Test and reflect

▼ **Extra practice · A revealing test**

REFERENCE · SECTION 8

Extra practice · A revealing test

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe **response** before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ Extra practice · Earlier transfer

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Reference · Selected W4 response copy

REFERENCE · SECTION 15

Reference · Selected W4 response copy

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

> **My reasoning and evidence**

▼ Extension · Graphs and honest scales — W1 Read the scale

REFERENCE · SECTION 17

Extension · Graphs and honest scales — W1 Read the scale

> Retained extension task with its own teacher key and response.

Extension question and record

- A bar is 6 squares long with 2 votes per square. How many votes is that?

My reasoning and evidence

▼ **Extension · Graphs and honest scales — W2 Compare fairly**

REFERENCE · SECTION 18

Extension · Graphs and honest scales — W2 Compare fairly

> Retained extension task with its own teacher key and response.

Extension question and record

- Counts are 12 and 15. State the difference and ratio using the true values.

My reasoning and evidence

▼ Extension · Graphs and honest scales — W3 Repair a misleading graph

REFERENCE · SECTION 19

Extension · Graphs and honest scales — W3 Repair a misleading graph

> Retained extension task with its own teacher key and response.

Extension question and record

- Bars for 12 and 15 start at 10. Explain the misleading lengths and repair.

> **My reasoning and evidence**

▼ Extension · Graphs and honest scales — W4 Draw independently

REFERENCE · SECTION 20

Extension · Graphs and honest scales — W4 Draw independently

> Retained extension task with its own teacher key and response.

Extension question and record

- Draw a **bar chart** for Music 6, Art 9, Coding 3. Include title, category labels, vote units, zero **baseline** and equal **scale**. State two supported comparisons.

> **My reasoning and evidence**

▼ **Extension · Graphs and honest scales — W5 Limit the claim**

REFERENCE · SECTION 21

Extension · Graphs and honest scales — W5 Limit the claim

> Retained extension task with its own teacher key and response.

Extension question and record

- Art has the most votes among 18 **response**s. Does that establish Art is best for learning?

> **My reasoning and evidence**

▼ Extra practice · Pictured case

REFERENCE · SECTION 23

Extra practice · Pictured case

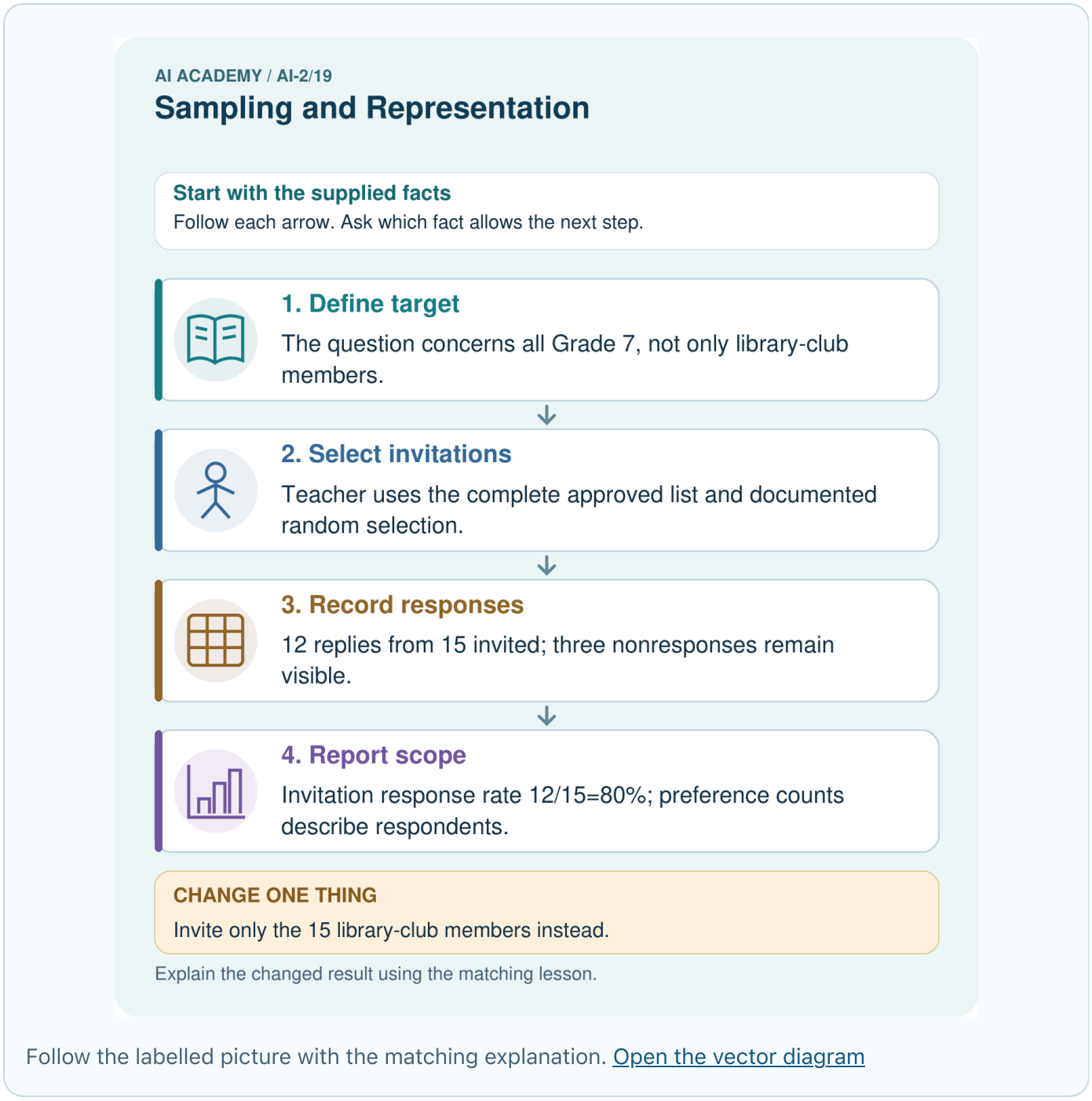
> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.



Follow the labelled picture with the matching explanation. Open the vector diagram

Can **80% response** be reported as **80%** of the whole grade responding?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

▼ **Reference · Project milestone**

REFERENCE · SECTION 24

Reference · Project milestone

> Use when helpful; keep this practice separate from the selected first attempt.

Additional evidence

Project: Transparent Book Recommender

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ Extra practice · Pictures, feedback and repair

REFERENCE · SECTION 25

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional Grade 7 class wants to understand book-format preferences across its whole grade.
- Which labelled step matches this detail? Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
- 1. Define target
- 2. Report scope
- 3. Record **response**s
- 4. Select invitations

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after trying**

After your attempt, compare with the supported explanation: Report scope Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Changed-case question

- Changed-case practice: Invite only the 15 library-club members instead. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The invitation **process** no longer covers the whole grade through this frame.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check changed case**

After your attempt, compare with the supported explanation: The invitation **process** no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Repair and keep evidence

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 26

Reference · Six-picture response sheets

Choose the relevant picture rather than repeating every task.

01. Sampling and Representation: the case

AI ACADEMY / AI-2/19 · 01 · LOOK AT THE FACTS

Sampling and Representation: the case

The situation

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Supplied evidence

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.
Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question.
No names enter the analysis table.

By the end, I can define a population, selection frame and nonresponse report for a survey.

Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Supplied facts: **Population**: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: The question concerns all Grade 7, not only library-club members. → Select invitations: Teacher uses the complete approved list and documented random selection. → Record **response**s: 12 replies from 15 invited; three non**response**s remain visible. → Report scope: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Follow the evidence

AI ACADEMY / AI-2/19 · 03 · INSPECT THE LESSON MODEL

Follow the evidence

Step or case	Evidence	Meaning
Target	All Grade 7 students	Population
Available data	Three club responses	Sample
Observed result	2 of 3 choose Code	Describes respondents

Read each row from left to right and connect the evidence to the stated rule.

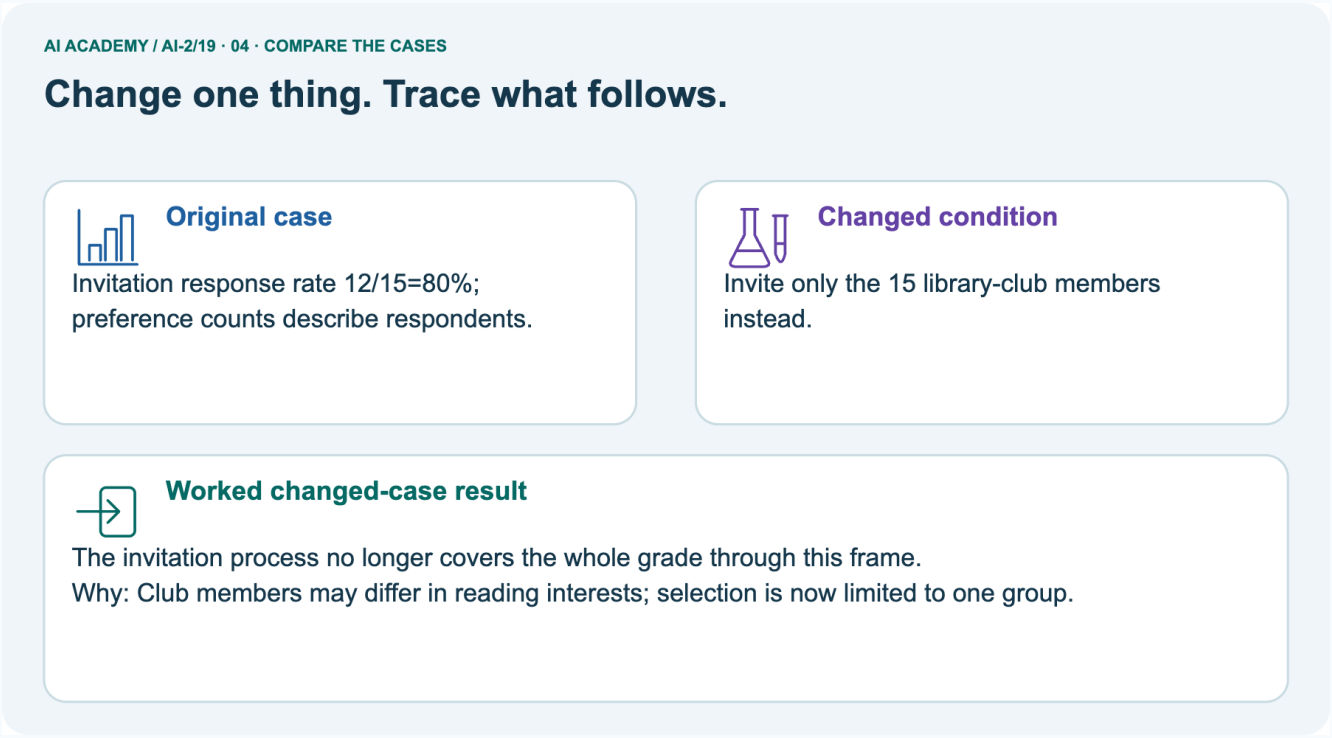
Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Target | All Grade 7 students | **Population**. Available data | Three club **response**s | **Sample**. Observed result | 2 of 3 choose Code | Describes respondents

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

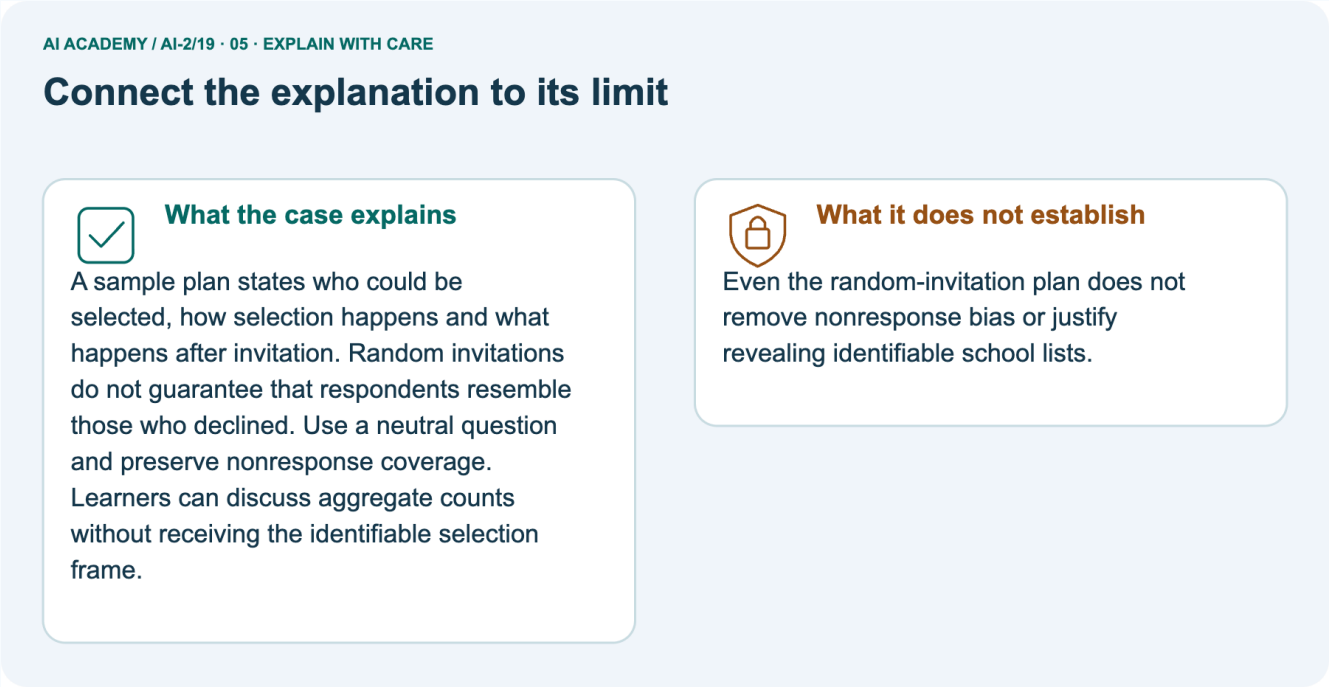
Original case: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
Changed condition: Invite only the 15 library-club members instead.

Expected result: The invitation **process** no longer covers the whole grade through this frame.
Reason: Club members may differ in reading interests; selection is now limited to one group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

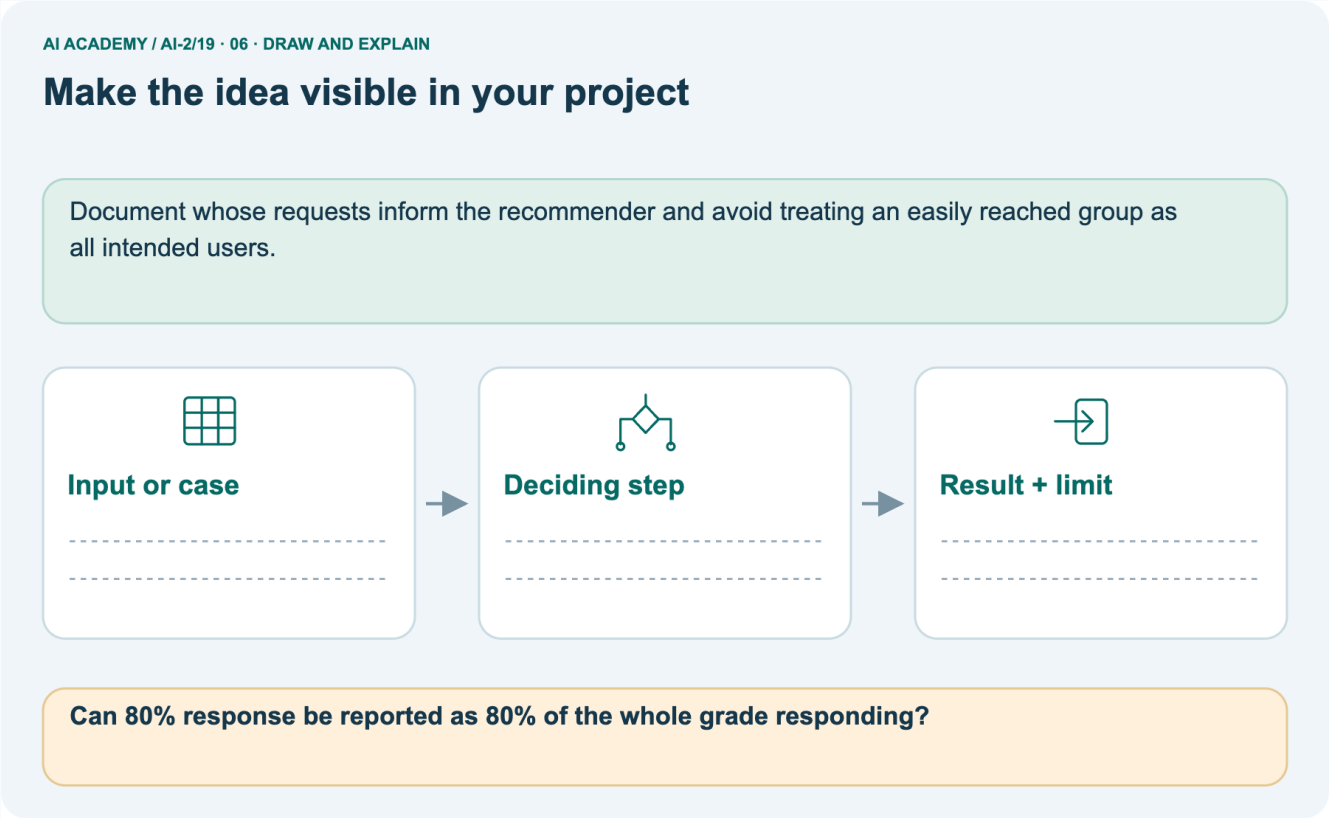
Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Limit: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users. Draw your **input**, deciding step, result and limit.

Supported question: Can **80% response** be reported as **80%** of the whole grade responding?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Sampling and Representation: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.